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Department of Electronics & Communication Engineering

LAB MANUAL

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AIM

To study the current–voltage (V–I) characteristics of a Gunn diode, to locate the threshold (peak) voltage, and to identify the negative–resistance region of the device.

EQUIPMENT AND COMPONENTS

- 1 Gunn power supply.
- 2 Gunn oscillator with Gunn diode mount.
- 3 PIN modulator.
- 4 Isolator.
- 5 Frequency meter.
- 6 Variable attenuator.
- 7 Detector mount with matched crystal detector.
- 8 Waveguide stands.
- 9 VSWR meter / Cathode Ray Oscilloscope.
- 10 BNC connecting cables and cooling fan.

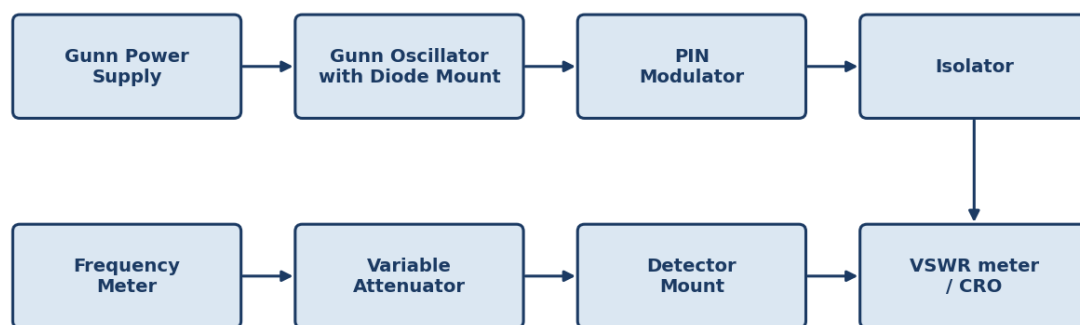
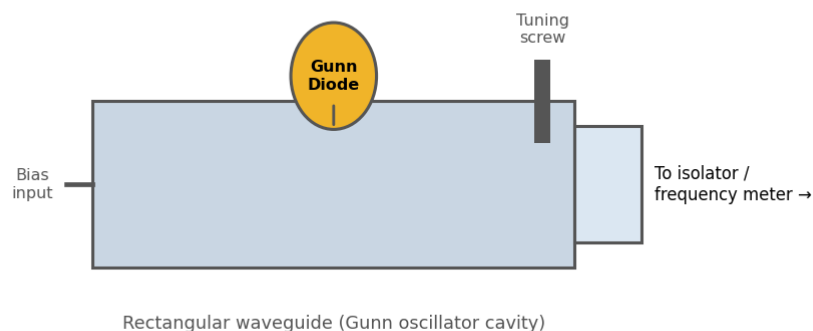
EXPERIMENTAL SETUP

Fig. 1 — Microwave bench arrangement for Gunn diode V–I characteristics

Gunn Oscillator with Diode Mount (schematic)

Photographic/schematic view of the Gunn oscillator with diode mount used in the bench

THEORY

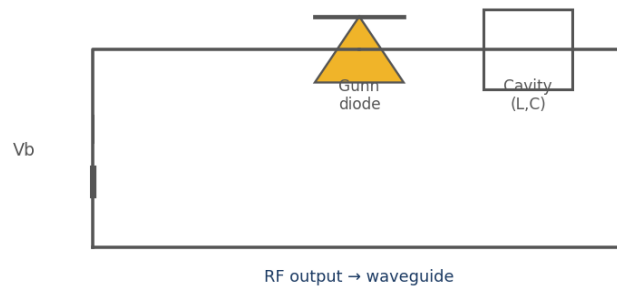
The Gunn diode is a two–terminal, negative–resistance active device made from a bulk semiconductor such as Gallium Arsenide (GaAs) or Indium Phosphide (InP), which does not use a p–n junction. Its operation is based on the Gunn effect, also called the transferred–electron effect. In these compound semiconductors, the conduction band has two energy valleys: a lower valley in which electrons have high mobility, and an upper satellite valley in which electrons have low mobility.

At low applied electric fields, nearly all the conduction electrons stay in the high-mobility lower valley and the current increases linearly with voltage, exhibiting normal ohmic behaviour. As the field is raised beyond a critical (threshold) value, a significant fraction of electrons gain enough energy to be scattered into the upper, low-mobility valley. Because their effective mobility now falls even though the field keeps increasing, the drift velocity – and hence the current – decreases with increasing voltage. This produces a region of negative differential resistance in the device characteristic. Beyond this region, high-field domains form and travel through the diode, giving rise to a periodic current oscillation whose frequency depends on the transit time of the domain across the active layer, which is the basis of Gunn oscillation in the microwave range.

The complete V–I curve of a Gunn diode therefore consists of three broad regions: an initial ohmic rise, a negative-resistance region beyond the threshold (peak) voltage where current falls as voltage rises, and a saturation region at higher voltages.

CIRCUIT / PORT DIAGRAM

Gunn Diode Bias Circuit (equivalent representation)

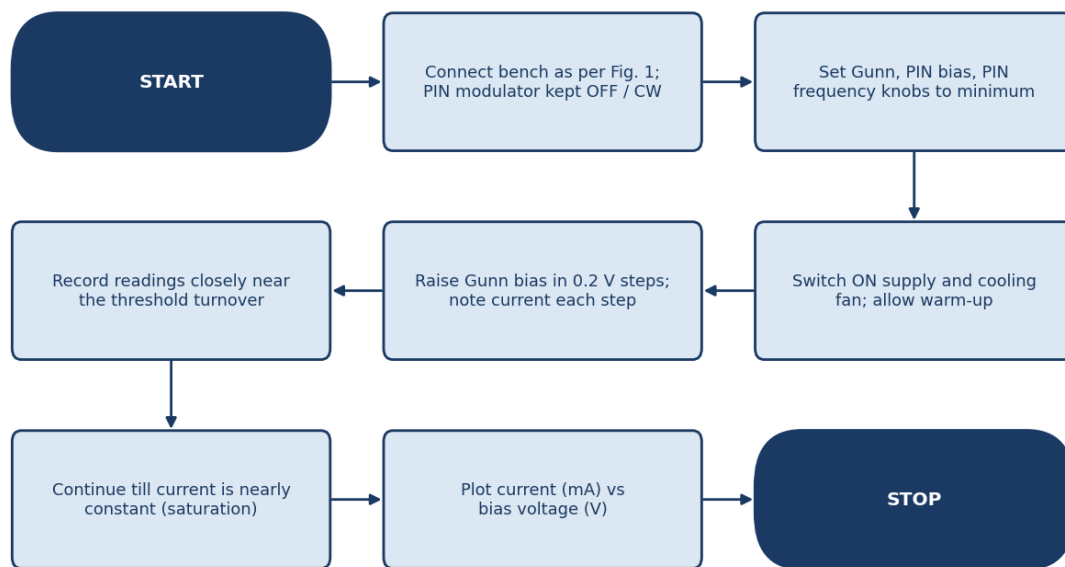


Equivalent bias circuit of the Gunn diode showing the negative-resistance element coupled to the resonant cavity

PROCEDURE

1. The equipment is connected as shown in Fig. 1, with the PIN modulator kept in the “OFF”/CW position so that only the diode bias is varied (no modulation is applied for this test).
2. All the control knobs on the Gunn power supply (Gunn bias, PIN bias, PIN frequency) are initially set to the minimum position.
3. The Gunn power supply and the cooling fan are switched ON, and a short warm-up time is allowed.
4. The Gunn bias voltage knob is increased in small steps (of about 0.2 V), and the corresponding meter current is noted at every step.
5. Readings are taken carefully around the region where the current stops rising and begins to fall, since this identifies the threshold voltage.
6. The bias voltage is increased further, in the same steps, until the current again becomes nearly constant (saturation), and readings are recorded throughout.
7. A graph of diode current (mA) versus bias voltage (V) is plotted from the tabulated readings.

PROCEDURE FLOWCHART



Flowchart — sequence of steps followed in Exp-01

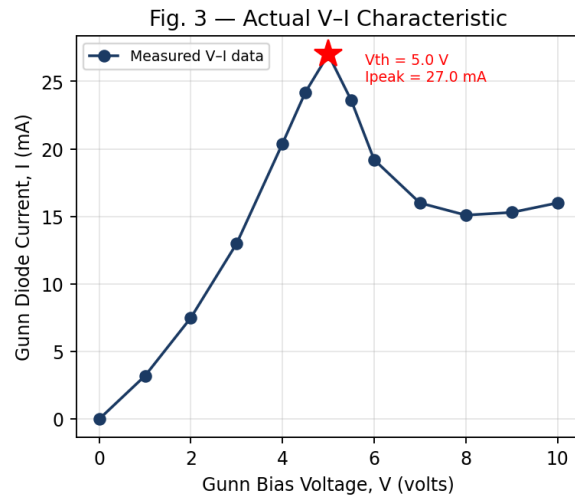
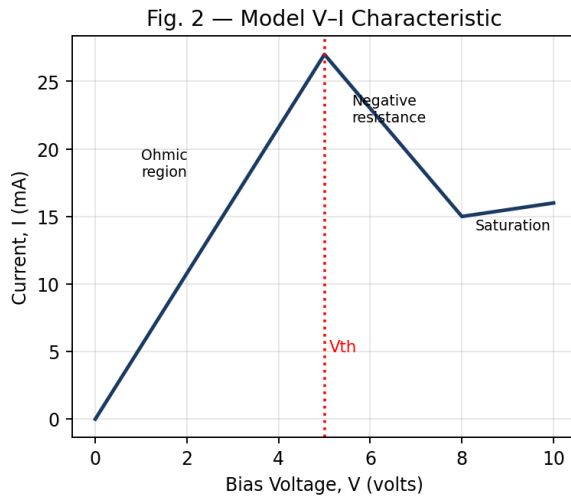
TABULATION

S. No.	Gunn Bias Voltage (V)	Gunn Diode Current (mA)
1	0.0	0.0
2	1.0	3.2
3	2.0	7.5
4	3.0	13.0
5	4.0	20.4
6	4.5	24.2
7	5.0 (V _{th})	27.0 (I _{th} , peak)
8	5.5	23.6
9	6.0	19.2
10	7.0	16.0
11	8.0	15.1
12	9.0	15.3
13	10.0	16.0

FORMULA

1. Threshold field, $E_{th} = V_{th} / d$, where d is the active-layer length of the diode (taken as $d \approx 15.2$ microns (μm) for this GaAs Gunn diode).

MODEL GRAPH AND ACTUAL GRAPH



CALCULATION

1. From the tabulated readings and the plotted graph, the current increases with bias voltage up to a maximum value of 27.0 mA at 5.0 V. Beyond this point the current falls with increasing voltage up to about 8.0 V, after which it levels off (saturation region).
2. Threshold voltage, $V_{th} = 5.0 \text{ V}$ (from graph, at $I_{peak} = 27.0 \text{ mA}$).
3. Negative-resistance region: $5.0 \text{ V} \leq V \leq 8.0 \text{ V}$ ($\Delta I / \Delta V < 0$ in this range).
4. $E_{th} = V_{th} / d = 5.0 \text{ V} / (15.2 \times 10^{-4} \text{ cm}) \approx 3.3 \times 10^3 \text{ V/cm} = 3.3 \text{ kV/cm}$, which agrees with the standard threshold field quoted for bulk GaAs.

PRECAUTIONS

1. All bias knobs must be kept at the minimum position before switching the power supply ON.
2. The Gunn bias voltage must never be increased beyond the safe operating limit specified for the diode (typically not more than about 10 V), to avoid damaging the device.
3. Readings should be taken only after allowing sufficient warm-up time.
4. Connections should be checked for tightness before switching ON.
5. Do not look directly into the open end of the waveguide.
6. The cooling fan must remain ON throughout the experiment.
7. After completing the experiment, all knobs should be returned to the minimum position before switching OFF.

RESULT / CONCLUSION

The V-I characteristics of the given Gunn diode were studied and plotted. The threshold (peak) voltage was found to be $V_{th} = 5.0 \text{ V}$, corresponding to a peak current of **27.0 mA**, and the device exhibited a clear negative-resistance region between **5.0 V and 8.0 V**, confirming the transferred-electron (Gunn) effect.

AIM

To study the mode characteristics of a reflex klystron and hence to determine the mode number, electron transit time, electronic tuning range (ETR) and electronic tuning sensitivity (ETS).

EQUIPMENT AND COMPONENTS

- 1 Klystron power supply.
- 2 Klystron tube (type 2K25) with mount.
- 3 Isolator.
- 4 Frequency meter.
- 5 Variable attenuator.
- 6 Detector mount.
- 7 Waveguide stands.
- 8 VSWR meter.
- 9 Cathode Ray Oscilloscope.

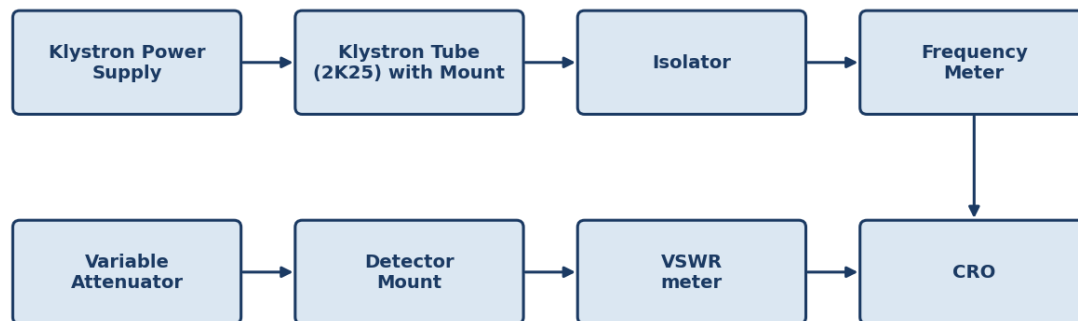
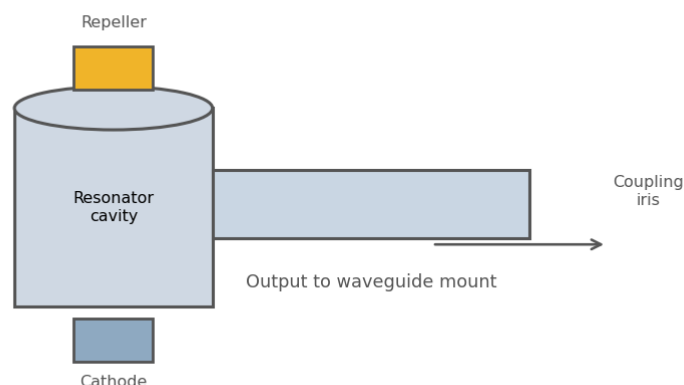
EXPERIMENTAL SETUP

Fig. 1 — Microwave bench setup for reflex klystron mode characteristics

Reflex Klystron Tube (2K25) with Mount (schematic)

Schematic view of the reflex klystron tube (2K25) with resonator, repeller and mount

THEORY

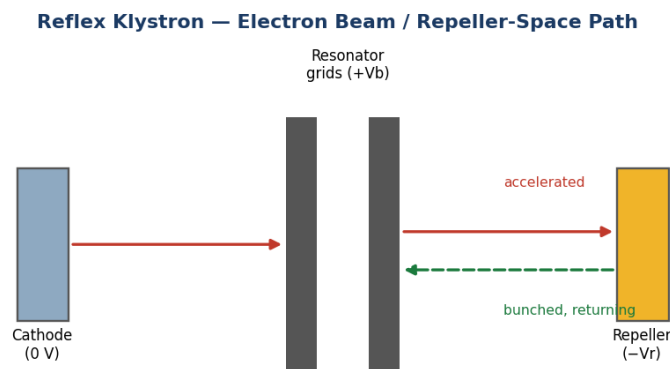
The reflex klystron is a low-power microwave source widely used in the laboratory. It generates microwave power using velocity modulation of an electron beam. Electrons emitted from the cathode are accelerated toward, and pass through, the positive resonator grids, after which they travel into the region beyond the resonator, called the repeller (reflector) space, where a repeller electrode is held at a negative potential with respect to the cathode. Because of this negative potential, the electrons are decelerated, brought to rest, and turned back (reflected) toward the resonator grids – hence the name “reflex” klystron.

While the beam is oscillating inside the resonator cavity, a radio–frequency field exists across the resonator gap. An electron that crosses the gap while the field is decelerating it loses energy and returns at a lower velocity, while an electron that crosses when the field is accelerating it gains energy and returns faster. Because of this difference, electrons that pass through the gap at different instants of the RF cycle take different times to travel into the repeller space and return – that is, they acquire different transit times. This is velocity modulation. As a result, on the return journey the electrons bunch together in groups (electron bunching).

If a bunch of returning electrons crosses the resonator gap at the instant when the gap field is decelerating (so that the bunch gives up energy to the cavity), the cavity is reinforced and oscillations build up and are sustained. This condition is met when the transit time in the repeller region corresponds to $(n + \frac{3}{4})$ cycles of the resonator frequency, where n is a positive integer or zero; each value of n identifies a distinct oscillation “mode” of the tube. If the bunch instead arrives when the field would accelerate it, energy is taken from the electrons but delivered to the field only if timed correctly – otherwise energy is removed from the resonator and oscillation is not sustained.

Because several different repeller voltages can satisfy the $(n + \frac{3}{4})$ transit–time condition, the tube oscillates over several discrete repeller–voltage ranges called modes, each with a characteristic output power and frequency. Tuning within a given mode by varying the repeller voltage is called electronic tuning; the corresponding shift in frequency per volt of repeller change is the electronic tuning sensitivity (ETS), and the total frequency range that can be covered electronically within a mode is the electronic tuning range (ETR). Separately, mechanical tuning changes the physical cavity dimensions, and hence its resonant frequency, without significantly changing the output power.

CIRCUIT / PORT DIAGRAM

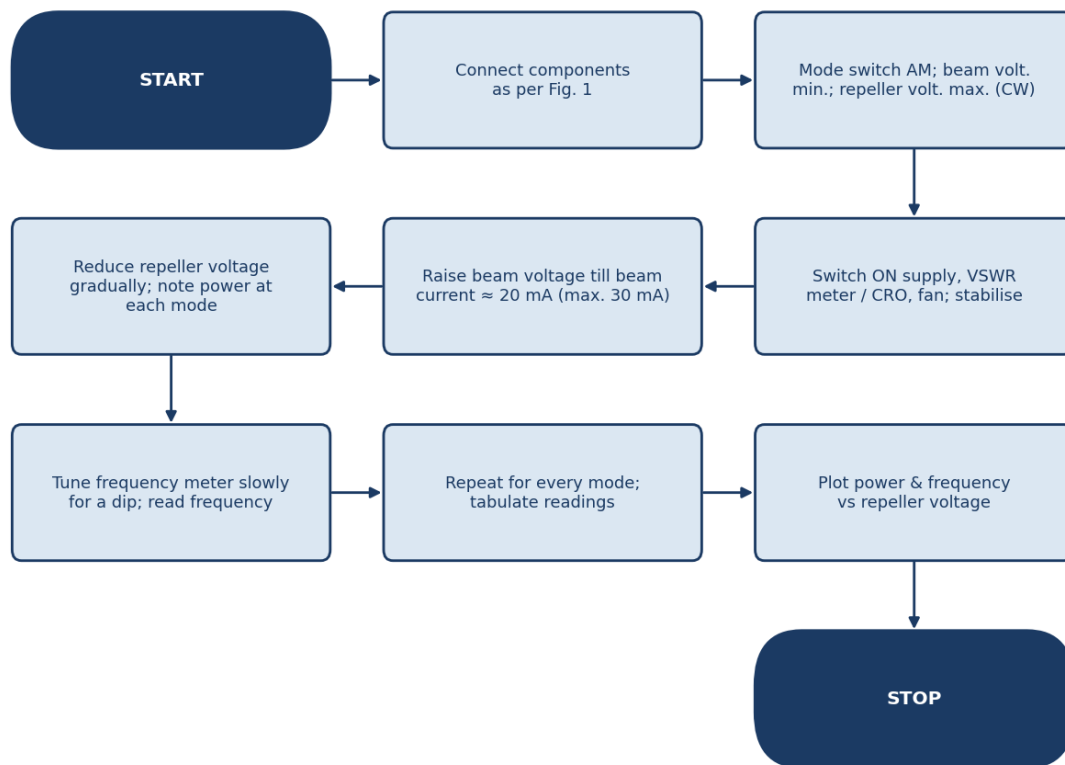


Electron beam path within the reflex klystron: acceleration through the resonator grids and bunched return from the repeller space

PROCEDURE

1. The components are connected as shown in Fig. 1.
2. Initial settings on the klystron power supply are made: mode switch to AM, beam voltage knob fully anticlockwise, repeller voltage knob fully clockwise, and the meter switch set to beam current.
3. The frequency meter is set off–resonance (turned to one side) and must always be rotated very slowly.
4. The klystron power supply, VSWR meter / CRO, and the cooling fan are switched ON, and about 1–2 minutes are allowed for the tube to stabilise.
5. The beam voltage is increased from minimum until the beam current is around 20 mA, ensuring it never exceeds 30 mA.
6. The meter switch is changed to the repeller voltage position, and the repeller voltage is reduced gradually from its maximum negative value; at each mode (region of oscillation) the detected output power and the corresponding repeller voltage are noted.
7. To measure the oscillation frequency at a given mode, the mode switch is set to AM, and the output is observed on the CRO; the frequency meter is rotated slowly until a sharp dip appears in the displayed output, and the frequency reading is taken at that dip.
8. The above steps are repeated for each visible mode, recording the repeller voltage, output power (via the VSWR meter) and frequency (via the dip on the frequency meter) for every mode.
9. A graph of output power versus repeller voltage is plotted to display the mode pattern, and a graph of frequency versus repeller voltage is plotted for each mode to obtain the electronic tuning characteristic.

PROCEDURE FLOWCHART



Flowchart — sequence of steps followed in Exp-02

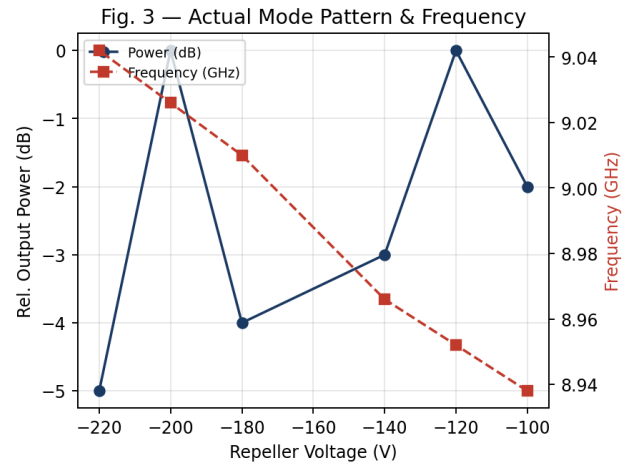
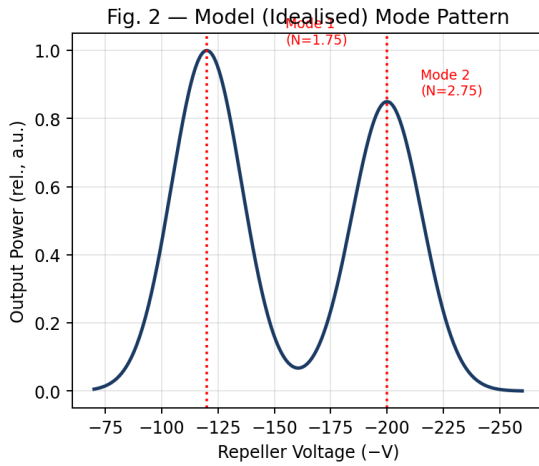
TABULATION

S. No.	Repeller Voltage (V)	Relative Output Power (dB)	Frequency (GHz)	Mode
1	-100	-2.0	8.938	Mode 1 (n = 1, N = 1.75)
2	-120	0.0 (peak)	8.952	
3	-140	-3.0	8.966	
4	-180	-4.0	9.010	Mode 2 (n = 2, N = 2.75)
5	-200	0.0 (peak)	9.026	
6	-220	-5.0	9.042	

FORMULA

1. Mode number: $N = n + \frac{3}{4}$ ($n = 0, 1, 2, \dots$)
2. Electronic Tuning Sensitivity: $ETS = (f_2 - f_1) / (V_2 - V_1)$ MHz/V
3. Electronic Tuning Range (ETR): $ETR = f_{\max} - f_{\min}$ within a mode (MHz)
4. Output power: $P_o = 10^{(x/20)}$ watts, where x is the dB reading on the VSWR meter

MODEL GRAPH AND ACTUAL GRAPH



CALCULATION

1. Mode 1 (peak at -120 V): $N_1 = n + \frac{3}{4} = 1.75$ ($n = 1$).
2. $ETS_1 = (f_{140} - f_{100}) / (V_{140} - V_{100}) = (8.966 - 8.938) \text{ GHz} / (-140 - (-100)) \text{ V} = 28 \text{ MHz} / (-40 \text{ V}) \approx 0.70 \text{ MHz/V}$.
3. $ETR_1 = 8.966 - 8.938 \text{ GHz} = 28 \text{ MHz}$.
4. Mode 2 (peak at -200 V): $N_2 = n + \frac{3}{4} = 2.75$ ($n = 2$).
5. $ETS_2 = (9.042 - 9.010) \text{ GHz} / (-220 - (-180)) \text{ V} = 32 \text{ MHz} / (-40 \text{ V}) \approx 0.80 \text{ MHz/V}$.
6. $ETR_2 = 9.042 - 9.010 \text{ GHz} = 32 \text{ MHz}$.

PRECAUTIONS

1. All connections should be checked before switching ON the power supply.
2. All knobs should be kept at their minimum position before switching ON the klystron / VSWR power supplies.
3. The klystron power supply "HT" switch must be OFF before the main supply is switched ON.
4. The beam voltage knob should be fully anticlockwise and the repeller voltage knob fully clockwise before starting.
5. Sufficient warm-up time should be given after switching ON the main supply.
6. Never look directly into the open waveguide.
7. At the end of the experiment, all knobs are returned to minimum before switching OFF the mains.
8. If the mains supply fails mid-experiment, all knobs must first be returned to minimum before the main switches are turned OFF.
9. The repeller voltage must be kept within -70 V to -270 V and never exceeded.

RESULT / CONCLUSION

The mode characteristics of the given reflex klystron (2K25) were studied over the range -70 V to -270 V. Two oscillation modes were observed, **$N_1 = 1.75$** (peak at -120 V, 8.952 GHz) and **$N_2 = 2.75$** (peak at -200 V, 9.026 GHz). The electronic tuning sensitivity was found to be **0.70 MHz/V** for Mode 1 and **0.80 MHz/V** for Mode 2, with electronic tuning ranges of **28 MHz** and **32 MHz** respectively.

AIM

To study a microwave directional coupler and to determine its coupling factor, insertion loss, isolation and directivity.

EQUIPMENT AND COMPONENTS

- 1 Klystron power supply.
- 2 Klystron tube (2K25) with mount.
- 3 Isolator.
- 4 Frequency meter.
- 5 Variable attenuator.
- 6 Directional coupler (device under test).
- 7 Two detector mounts.
- 8 Matched termination.
- 9 VSWR meter.
- 10 Waveguide stands.

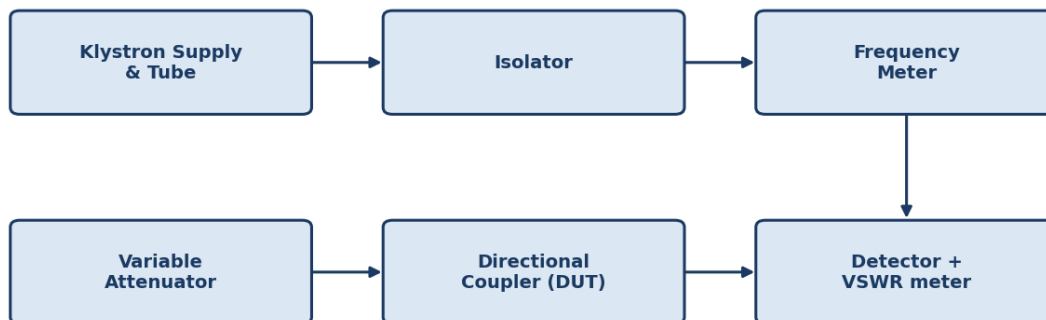
EXPERIMENTAL SETUP

Fig. 1 — Microwave bench setup for directional coupler measurements

THEORY

A directional coupler is a four-port waveguide junction consisting of a primary (main) waveguide and a secondary (auxiliary) waveguide coupled to it through one or more small apertures. In a typical two-hole (Bethe-hole type) coupler, the coupling holes are spaced a quarter of a guide wavelength apart, so that the waves coupled through the two holes add constructively in the forward direction of the secondary guide and cancel in the reverse direction. This gives the device its directional property: power injected at the input port mainly couples into the secondary guide travelling toward the “coupled” port, while very little couples toward the “isolated” port.

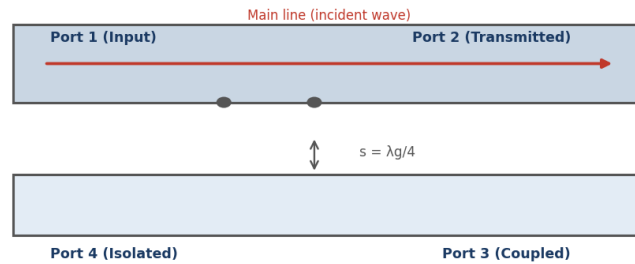
The four ports of a directional coupler are usually identified as: the input port, the transmitted (through) port on the main line, the coupled port on the secondary line (in the forward direction), and the isolated port on the secondary line (in the reverse direction, ideally carrying no power). For a two-hole coupler the holes are separated by $s = \lambda_g/4$, where λ_g is the guide wavelength at the operating frequency; this quarter-wave spacing is what makes the two coupled contributions add in the forward direction and cancel in the reverse direction.

All readings in this experiment are taken directly in dB on a calibrated VSWR meter, which uses a square-law crystal detector followed by a logarithmic amplifier; hence the meter reading is already proportional to power in dB and no separate detector-law correction is required for the calculations below.

The performance of a directional coupler is described by the following parameters, all expressed in decibels: the coupling factor, which is the ratio of the input power to the coupled power; the insertion loss, the ratio of input power to the power delivered at the through port; the isolation, the ratio of input power to the power leaking to the isolated port; and the directivity, the ratio of the coupled power to the power at the isolated port, which is a measure of how well the coupler distinguishes the direction of the travelling wave.

CIRCUIT / PORT DIAGRAM

Two-Hole Directional Coupler — Port Layout

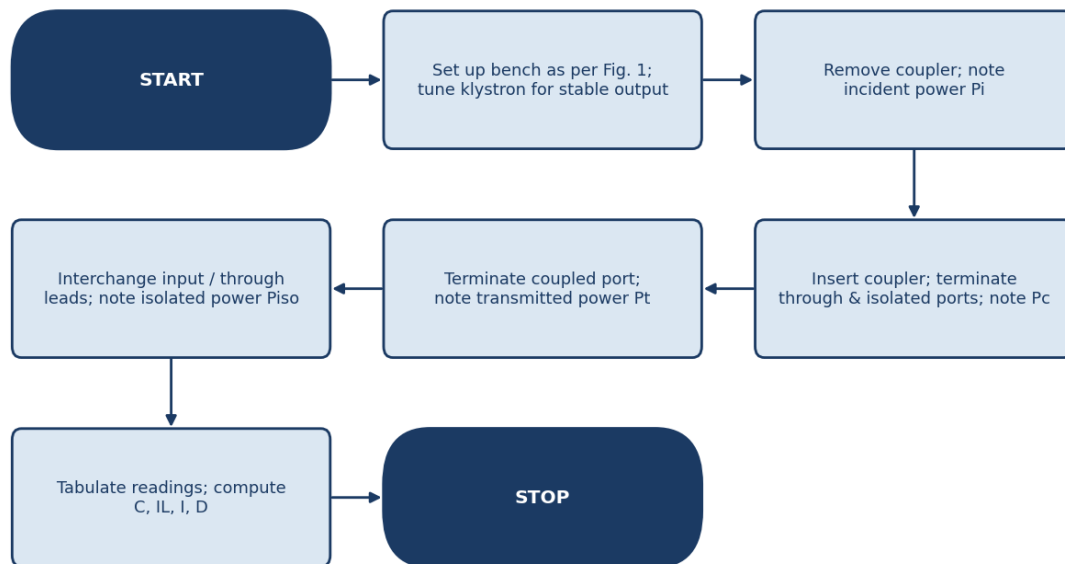


Two-hole directional coupler showing the four ports and the quarter-wavelength coupling-hole spacing

PROCEDURE

1. The bench is set up as in Fig. 1 with the klystron tuned to give a stable output at the working frequency.
2. With the directional coupler removed, the detector is connected directly after the variable attenuator and the incident power P_i is noted on the VSWR meter.
3. Coupling factor: the coupler is inserted, the through port is terminated in a matched load, the isolated port is also terminated, and the detector is connected to the coupled port; the reading P_c is noted.
4. Insertion loss: with the coupled port terminated in a matched load, the detector is connected to the through port and the transmitted power P_t is noted.
5. Isolation / Directivity: the input and through connections are interchanged (power fed from the opposite end / through port terminated appropriately) so that power now travels toward the isolated port; the detector reading P_{iso} at the isolated port is noted with the coupled port similarly excited for comparison.
6. All readings are recorded in dB (or converted to power) and used to compute the four parameters using the formulae below.

PROCEDURE FLOWCHART



Flowchart — sequence of steps followed in Exp-03

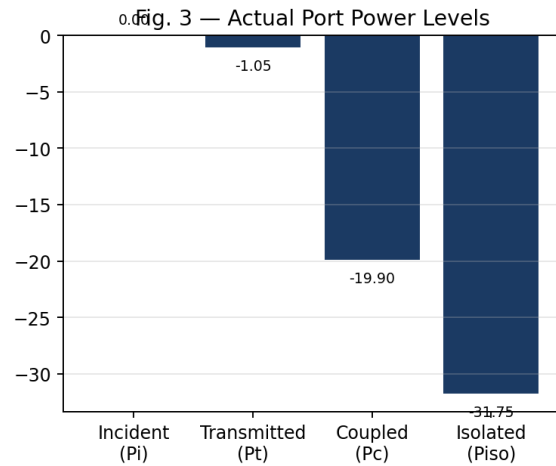
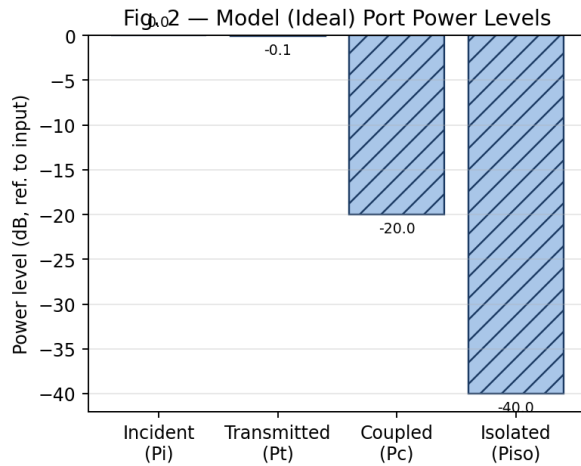
TABULATION

Trial	Incident P_i (dB)	Transmitted P_t (dB)	Coupled P_c (dB)	Isolated P_{iso} (dB)
1	0.0 (ref.)	-1.0	-20.0	-32.0
2	0.0 (ref.)	-1.1	-19.8	-31.5
Average	0.0	-1.05	-19.9	-31.75

FORMULA

1. Coupling Factor, $C = P_i - P_c = 10 \log_{10} (P_i / P_c)$ dB
2. Insertion Loss, $IL = P_i - P_t = 10 \log_{10} (P_i / P_t)$ dB
3. Isolation, $I = P_i - P_{iso} = 10 \log_{10} (P_i / P_{iso})$ dB
4. Directivity, $D = I - C$ dB

MODEL GRAPH AND ACTUAL GRAPH



CALCULATION

1. Using the average readings (P_i taken as the 0 dB reference):
2. $C = 0.0 - (-19.9) = 19.9$ dB
3. $IL = 0.0 - (-1.05) = 1.05$ dB
4. $I = 0.0 - (-31.75) = 31.75$ dB
5. $D = I - C = 31.75 - 19.9 = 11.85$ dB

PRECAUTIONS

1. All ports not being measured must be terminated in matched loads to avoid reflections that would distort the readings.
2. Connections between waveguide flanges should be tight and properly aligned.
3. The klystron output should be kept stable (constant beam and repeller voltage) throughout a set of readings.
4. The frequency meter should be de-tuned (off resonance) except when measuring frequency, to avoid an additional loss.
5. Do not look directly into the open waveguide.

RESULT / CONCLUSION

The characteristics of the given microwave directional coupler were measured. Coupling Factor **C = 19.9 dB** (close to its rated 20 dB coupling), Insertion Loss **IL = 1.05 dB**, Isolation **I = 31.75 dB**, and Directivity **D = 11.85 dB**, values that are consistent with a typical two-hole X-band waveguide directional coupler.

AIM

To study the characteristics of a magic tee (hybrid junction) and to determine the isolation and the coupling coefficients between its arms.

EQUIPMENT AND COMPONENTS

- 1 Klystron power supply.
- 2 Klystron tube (2K25) with mount.
- 3 Isolator.
- 4 Frequency meter.
- 5 Variable attenuator.
- 6 Magic tee (device under test).
- 7 Two matched terminations.
- 8 Detector mount.
- 9 VSWR meter.
- 10 Waveguide stands.

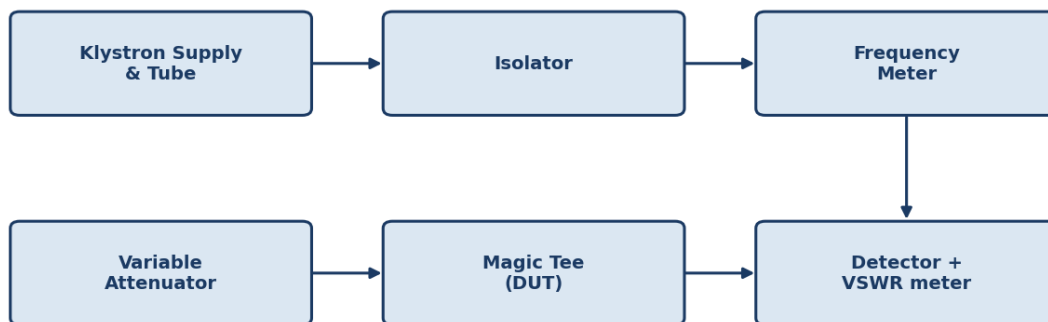
EXPERIMENTAL SETUP

Fig. 1 — Microwave bench setup with magic tee as device under test

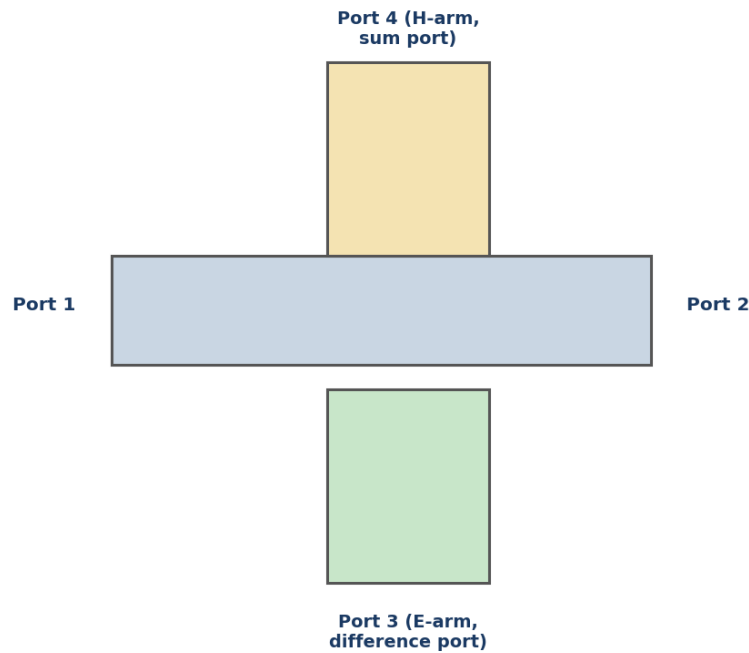
THEORY

A magic tee is a four-port waveguide junction formed by combining an E-plane tee and an H-plane tee at the same point of a main waveguide. It has two collinear arms (ports 1 and 2), an E-arm (port 3, also called the difference or sum-minus port) and an H-arm (port 4, also called the sum port). The key property of the magic tee is that the E-arm and the H-arm are isolated from each other – ideally, power fed into one of these arms does not appear at the other, provided the two collinear arms are terminated identically.

If power is fed into port 1 (a collinear arm), it divides equally between ports 3 and 4, with the two output signals in phase with each other in specific pairs depending on which collinear arm is excited. If power is fed into the E-arm (port 3), it splits equally between the two collinear arms but out of phase by 180°; if fed into the H-arm (port 4), it splits equally and in phase between the two collinear arms. Because of these properties, the magic tee is used in balanced mixers, microwave bridges and duplexers.

For an ideal, perfectly matched magic tee (with ports ordered 1, 2, 3(E), 4(H)), the scattering matrix has zero diagonal terms (perfect match at every port) and zero S_{34} / S_{43} terms (perfect isolation between the E- and H-arms):

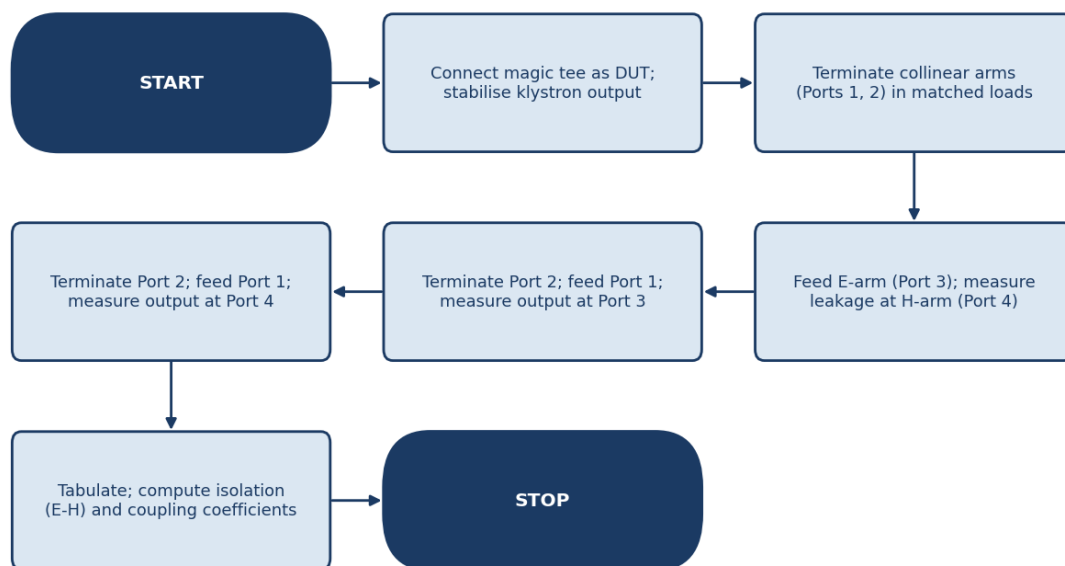
$$S = 1/\sqrt{2} \times \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & -1 \\ 1 & 1 & 0 & 0 \\ 1 & -1 & 0 & 0 \end{bmatrix}$$

Magic Tee — Four-Port Hybrid Junction

Magic tee — four-port hybrid junction: collinear arms (1, 2), H-arm (4, sum port), E-arm (3, difference port)

PROCEDURE

1. The bench is set up with the magic tee connected as the device under test, and the klystron output is stabilised.
2. Isolation between E and H arms: matched loads are connected to the two collinear arms (ports 1 and 2); power is fed into the E–arm (port 3) and the detector is connected to the H–arm (port 4). The leakage reading is noted and compared with the input power to obtain the isolation.
3. Coupling coefficients: with port 2 terminated in a matched load, power is fed into port 1 and the output is measured successively at port 3 and port 4 (with the other left correctly terminated), to find the power division from a collinear arm into the E– and H–arms.
4. All readings are taken on the VSWR meter (in dB) and tabulated.

PROCEDURE FLOWCHART

Flowchart — sequence of steps followed in Exp-04

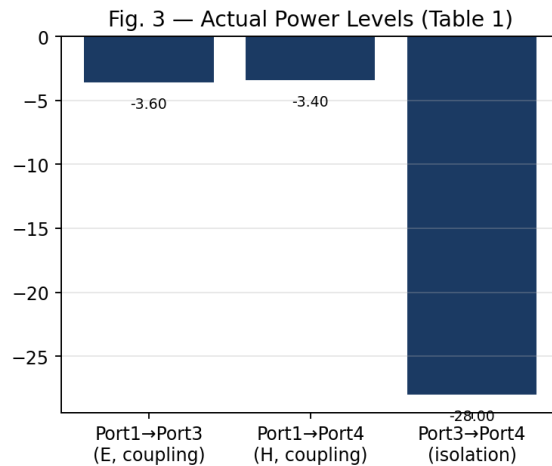
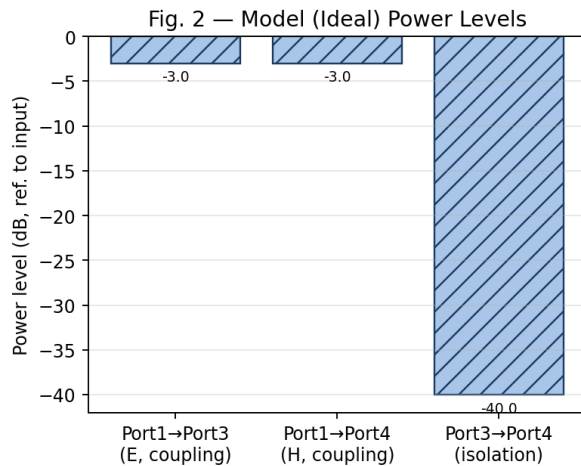
TABULATION

S. No.	Input Port	Output Port	Input Power (dB)	Output Power (dB)	Result (dB)
1	Port 3 (E)	Port 4 (H)	0.0	-28.0	Isolation = 28.0
2	Port 1	Port 3 (E)	0.0	-3.6	Coupling = 3.6
3	Port 1	Port 4 (H)	0.0	-3.4	Coupling = 3.4

FORMULA

1. Isolation (E – H) = $P_{in} - P_{out}$ dB
2. Coupling coefficient (port i → port j) = $P_i - P_j$ dB

MODEL GRAPH AND ACTUAL GRAPH



CALCULATION

1. Isolation (Port 3 → Port 4) = $0.0 - (-28.0) = 28.0$ dB
2. Coupling coefficient (Port 1 → Port 3) = $0.0 - (-3.6) = 3.6$ dB (close to the ideal 3.0 dB equal split)
3. Coupling coefficient (Port 1 → Port 4) = $0.0 - (-3.4) = 3.4$ dB

PRECAUTIONS

1. The two collinear arm ports must be terminated in identical, well-matched loads when measuring isolation, or the result will be inaccurate.
2. Waveguide flange connections must be tight and correctly aligned.
3. The klystron output should be kept stable while a set of readings is being taken.
4. Do not look directly into the open waveguide.

RESULT / CONCLUSION

The isolation between the E-arm and H-arm of the given magic tee was found to be **28.0 dB**, and the coupling coefficients from a collinear arm to the E- and H-arms were found to be **3.6 dB and 3.4 dB** respectively – close to the ideal 3 dB equal split predicted by theory, confirming correct operation of the junction.

AIM

To study the characteristics of an H-plane tee (shunt tee) and to determine the power-division ratio between its collinear arms and the isolation between them.

EQUIPMENT AND COMPONENTS

- 1 Klystron power supply.
- 2 Klystron tube (2K25) with mount.
- 3 Isolator.
- 4 Frequency meter.
- 5 Variable attenuator.
- 6 H-plane tee (device under test).
- 7 Matched terminations.
- 8 Detector mount.
- 9 VSWR meter.
- 10 Waveguide stands.

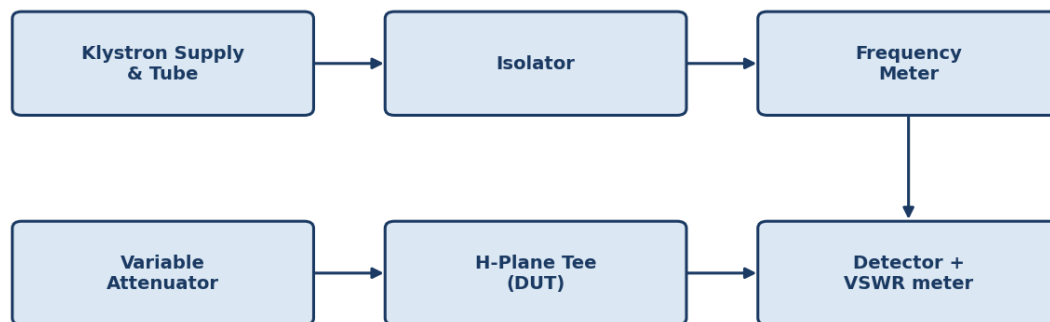
EXPERIMENTAL SETUP

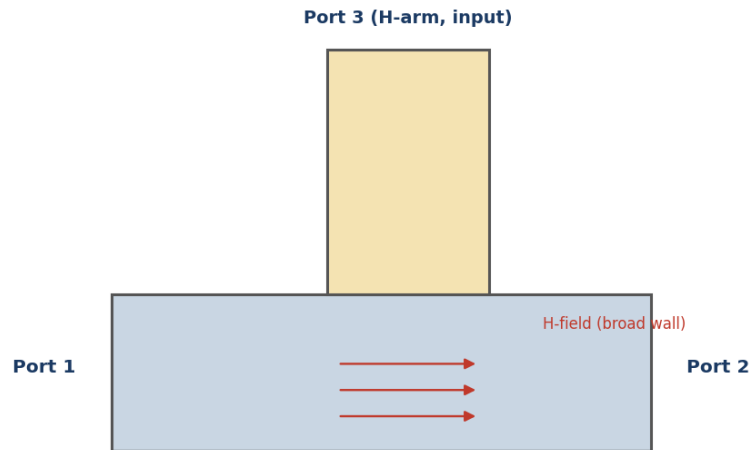
Fig. 1 — Microwave bench setup with H-plane tee as device under test

THEORY

An H-plane tee is a waveguide junction in which the axis of the side arm is parallel to the plane of the magnetic (H) field of the main waveguide, obtained by cutting a rectangular slot along the broad wall of the guide. It is a shunt junction: when power is fed into the side arm (port 3), it splits equally between the two collinear arms (ports 1 and 2) and emerges in phase at both. Conversely, when equal and in-phase signals are fed into the two collinear arms, they add constructively at the side arm; when fed with equal but out-of-phase signals, they cancel at the side arm. Because of the equal, in-phase power split it provides, the H-plane tee is commonly used as a power divider/combiner.

CIRCUIT / PORT DIAGRAM

H-Plane Tee (Shunt Junction) — Port Layout

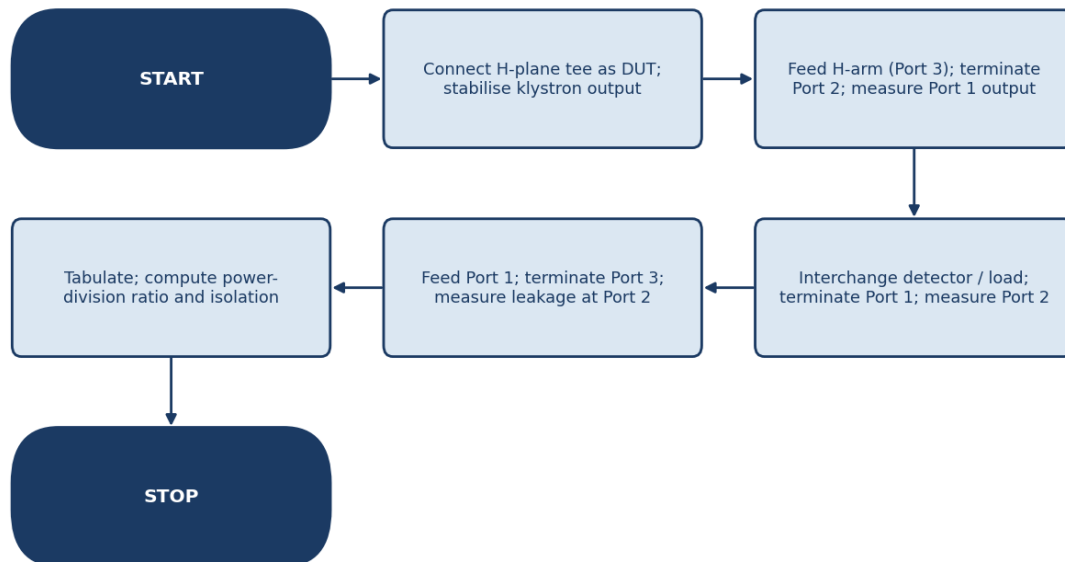


H-plane tee — side arm parallel to the H-field of the main guide (shunt junction)

PROCEDURE

1. The bench is set up with the H-plane tee connected as the device under test and the klystron output stabilised.
2. Power is fed into the H-arm (port 3); port 2 is terminated in a matched load and the detector is connected to port 1, and the output reading is noted.
3. The detector and matched load are then interchanged between ports 1 and 2, and the output at port 2 is noted with port 1 terminated.
4. To find the isolation between the two collinear arms, power is fed into port 1, port 3 is terminated in a matched load, and the leakage at port 2 is measured.
5. All readings are recorded in dB.

PROCEDURE FLOWCHART



Flowchart — sequence of steps followed in Exp-05

TABULATION

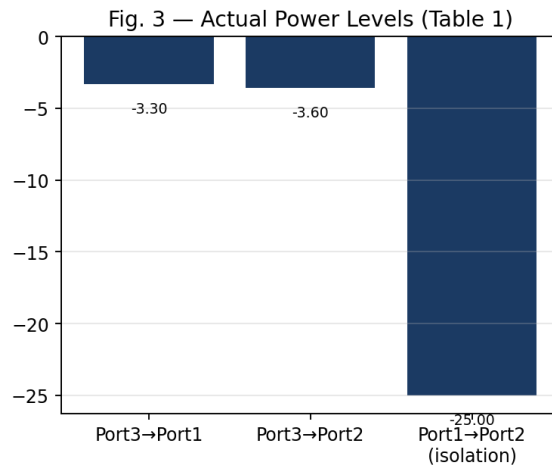
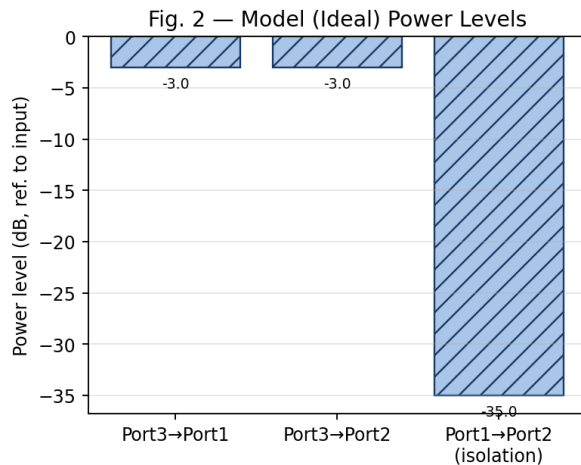
S. No.	Input Port	Output Port	Input Power (dB)	Output Power (dB)	Result (dB)
1	Port 3	Port 1	0.0	−3.3	3.3
2	Port 3	Port 2	0.0	−3.6	3.6

S. No.	Input Port	Output Port	Input Power (dB)	Output Power (dB)	Result (dB)
3	Port 1	Port 2 (isolation)	0.0	-25.0	25.0

FORMULA

1. Power division ratio (Port 3 → Port 1, Port 2) = $P_{in} - P_{out} = 10 \log_{10} (P_{in} / P_{out})$ dB (ideally ≈ 3 dB at each arm)
2. Isolation between collinear arms = $10 \log_{10} (P_1 / P_2)$ dB

MODEL GRAPH AND ACTUAL GRAPH



CALCULATION

1. Power at Port 1 (from Port 3) = $0.0 - (-3.3) = 3.3$ dB down.
2. Power at Port 2 (from Port 3) = $0.0 - (-3.6) = 3.6$ dB down.
3. Since both arms are within about 0.3 dB of each other and close to the ideal 3 dB point, the H-arm divides power essentially equally between the two collinear arms.
4. Isolation (Port 1 → Port 2, with Port 3 matched) = $0.0 - (-25.0) = 25.0$ dB.

PRECAUTIONS

1. Unused ports must always be terminated in matched loads.
2. Flange connections should be tight and properly aligned to avoid reflections.
3. The klystron source should be allowed to stabilise before readings are taken.
4. Do not look directly into the open waveguide.

RESULT / CONCLUSION

The power-division ratio between the collinear arms of the given H-plane tee was found to be **3.3 dB and 3.6 dB** (nearly equal, in phase), and the isolation between the collinear arms was found to be **25.0 dB**, confirming the equal in-phase power split expected of an H-plane (shunt) junction.

AIM

To study the characteristics of an E-plane tee (series tee) and to determine the power-division ratio between its collinear arms and the isolation between them.

EQUIPMENT AND COMPONENTS

- 1 Klystron power supply.
- 2 Klystron tube (2K25) with mount.
- 3 Isolator.
- 4 Frequency meter.
- 5 Variable attenuator.
- 6 E-plane tee (device under test).
- 7 Matched terminations.
- 8 Detector mount.
- 9 VSWR meter.
- 10 Waveguide stands.

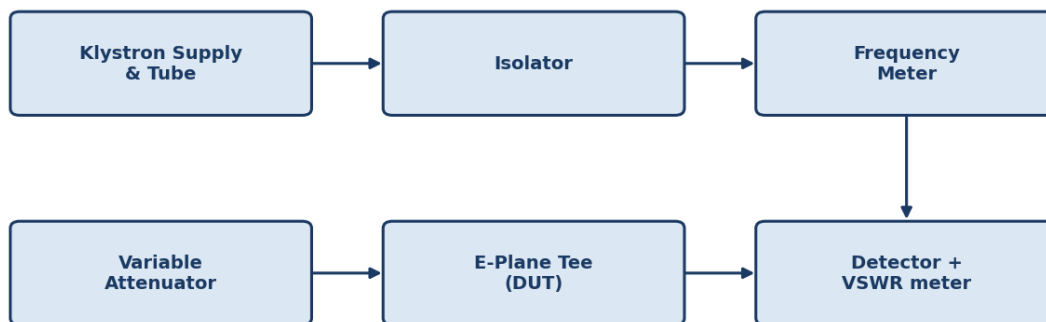
EXPERIMENTAL SETUP

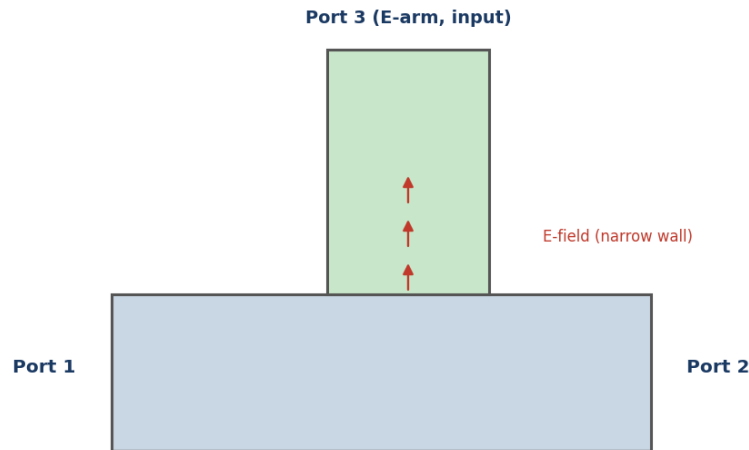
Fig. 1 — Microwave bench setup with E-plane tee as device under test

THEORY

An E-plane tee is a waveguide junction in which the axis of the side arm is parallel to the electric (E) field of the main waveguide, formed by cutting a slot along the narrow wall so the side arm extends in the plane containing the E-field. It behaves as a series junction: when power is fed into the side arm (port 3, the E-arm), it divides equally between the two collinear arms (ports 1 and 2) but the two outputs are 180° out of phase with each other. Conversely, signals fed into the two collinear arms in phase cancel at the E-arm, while signals fed 180° out of phase add at the E-arm. This phase-reversing power split makes the E-plane tee useful in balanced mixers and other circuits requiring a phase difference between the two output signals.

CIRCUIT / PORT DIAGRAM

E-Plane Tee (Series Junction) — Port Layout

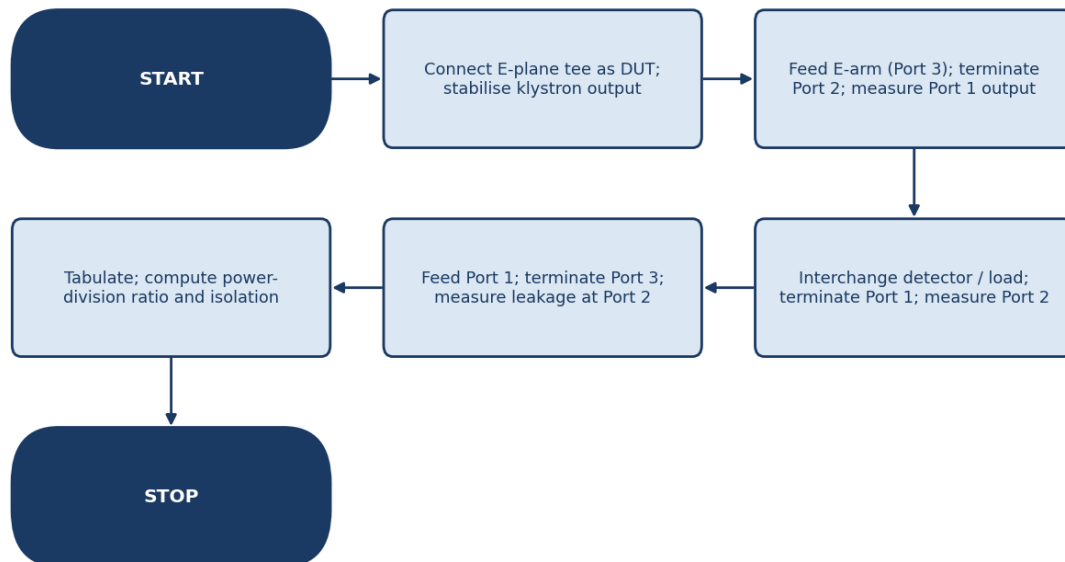


E-plane tee — side arm parallel to the E-field of the main guide (series junction)

PROCEDURE

1. The bench is set up with the E–plane tee connected as the device under test and the klystron output stabilised.
2. Power is fed into the E–arm (port 3); port 2 is terminated in a matched load and the detector is connected to port 1, and the output reading is noted.
3. The detector and matched load are interchanged between ports 1 and 2, and the output at port 2 is noted with port 1 terminated.
4. To find the isolation between the two collinear arms, power is fed into port 1, port 3 is terminated in a matched load, and the leakage at port 2 is measured.
5. All readings are recorded in dB.

PROCEDURE FLOWCHART



Flowchart — sequence of steps followed in Exp–06

TABULATION

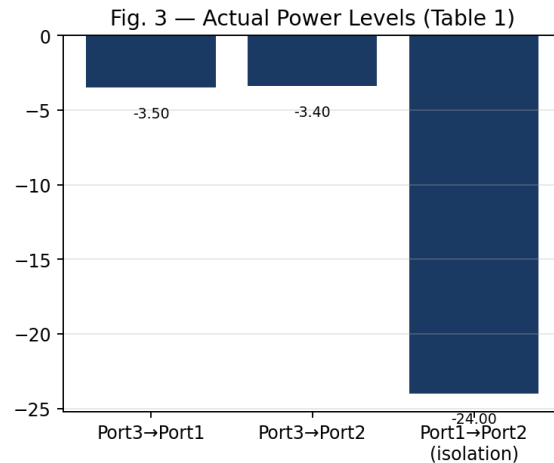
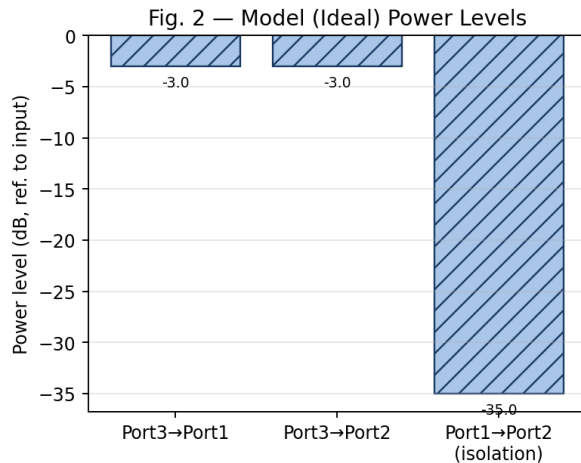
S. No.	Input Port	Output Port	Input Power (dB)	Output Power (dB)	Result (dB)
1	Port 3	Port 1	0.0	–3.5	3.5
2	Port 3	Port 2	0.0	–3.4	3.4

S. No.	Input Port	Output Port	Input Power (dB)	Output Power (dB)	Result (dB)
3	Port 1	Port 2 (isolation)	0.0	-24.0	24.0

FORMULA

1. Power division ratio (Port 3 → Port 1, Port 2) = $P_{in} - P_{out} = 10 \log_{10} (P_{in} / P_{out})$ dB (ideally ≈ 3 dB at each arm, 180° out of phase)
2. Isolation between collinear arms = $10 \log_{10} (P_1 / P_2)$ dB

MODEL GRAPH AND ACTUAL GRAPH



CALCULATION

1. Power at Port 1 (from Port 3) = $0.0 - (-3.5) = 3.5$ dB down.
2. Power at Port 2 (from Port 3) = $0.0 - (-3.4) = 3.4$ dB down.
3. Both arms are within 0.1 dB of each other and close to the ideal 3 dB point, confirming an equal power split; the 180° phase reversal between the two outputs (a property of the series junction) is confirmed separately using a phase-shifter/slotted-line comparison.
4. Isolation (Port 1 → Port 2, with Port 3 matched) = $0.0 - (-24.0) = 24.0$ dB.

PRECAUTIONS

1. Unused ports must always be terminated in matched loads.
2. Flange connections should be tight and properly aligned to avoid reflections.
3. The klystron source should be allowed to stabilise before readings are taken.
4. Do not look directly into the open waveguide.

RESULT / CONCLUSION

The power-division ratio between the collinear arms of the given E-plane tee was found to be **3.5 dB and 3.4 dB** (nearly equal), and the isolation between the collinear arms was found to be **24.0 dB**. The equal power split with a 180° phase reversal, characteristic of a series (E-plane) junction, was verified.